

SimVille: Exploring Social Complexity and Emotional Tone in LLM-Driven Virtual Reality Digital Health Game

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Figure 1: Teaser.

ABSTRACT

We present SimVille, a mixed reality game that combines extended reality (XR), large language models (LLMs), and conversational non-player characters (NPCs) to explore new forms of immersive, socially interactive learning of conversational skills. In SimVille, players engage in quests by conversing naturally with embodied NPCs, where dialogue is dynamically generated through LLMs. Beyond quest completion, the system investigates how adaptive dialogue, social cues, and narrative-driven embodiment can support the development of communication skills, such as small talk, irony detection, or emotion recognition. We describe the design of SimVille, including technical integration of XR and LLM pipelines, interaction mechanics, and a framework for evaluating conversational experiences. Our contribution examines the usability of AI agents in a virtual reality setting and demonstrates how adjustable difficulty levels can create a personalized experience for every user.

Results show that ...

Index Terms: Autism, Game, Large Language Models, Conversation, Generative AI, Virtual Reality, Game Complexity

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1 INTRODUCTION

Recent advances in extended reality (XR) and large language models (LLMs) are transforming how humans interact with artificial agents. As conversational models become capable of generating contextually appropriate and affectively aligned dialog, they enable novel forms of social simulation and training in immersive environments. The immersive environment reduces unwanted external stimuli and directs the user’s attention toward the virtual setting [26]. LLMs-powered virtual agents can respond dynamically, exhibit empathy, and adapt their communication style to user behavior—opening new opportunities for education, therapy, and social skills development.

Within this context, *SimVille* investigates how naturalistic, emotionally expressive virtual agents can support conversational learning in a socially rich XR setting. The system integrates immersive virtual reality (VR) with adaptive LLM-driven dialogue to simulate cooperative as well as challenging social encounters. Such experiences are particularly relevant for users who struggle with social communication, including individuals on the autism spectrum, as prior work shows that well-designed immersive environments can enhance social understanding and emotion regulation when tailored to cognitive and sensory needs [17, 12]. VR provides an immersive yet fully controlled environment in which users can safely practice real-world social interactions. In *SimVille*, this enables participants to engage in realistic conversations, perceive emotional cues, and receive adaptive feedback without fear of negative judgment.

Emotional tone plays a central role in shaping the cognitive and affective demands of social interaction. Prior research shows that negatively valenced or conflicting emotional cues can increase

conversational load, elevate stress responses, and require greater perspective-taking during human - agent interaction [20]. From a cognitive standpoint, users must allocate additional resources for emotion recognition, regulation, and Theory of Mind (ToM) reasoning when interlocutors display impatience, sarcasm, or frustration. Despite the growing use of conversational agents in VR, most existing systems rely on scripted or rule-based NPCs with fixed dialogue patterns, limiting the expressiveness and variability of social encounters. Our work addresses this gap by examining how an NPC's emotional tone, in combination with different levels of social complexity, shapes user experience, workload, and engagement in an immersive conversational setting.

The study investigates how the number of conversational partners and their emotional tone influence user engagement, comfort, and cognitive load in a virtual restaurant scenario. Two research questions guide the experiment:

1. **RQ1:** How does the emotional tone of NPCs influence users' emotional experience and perceived cognitive load?
2. **RQ2:** How does increasing social complexity affect user experience across different social interaction scenarios?

Contributions. This work makes the following contributions to the study of social interaction with LLM-driven agents in immersive virtual reality:

- **A mixed-reality conversational system (SimVille)** that integrates real-time speech recognition, large language model (LLM) dialogue generation, and embodied non-player characters (NPCs) with controllable emotional tone.
- **A prompt and behavior framework** enabling fine-grained modulation of NPC emotional expressions (friendly vs. unfriendly) and structured conversational challenges within a VR restaurant scenario.
- **A controlled experimental design** implementing a 2×2 manipulation of social complexity (one vs. five NPCs) and emotional climate (friendly vs. unfriendly), allowing systematic investigation of conversational load in LLM-driven VR.
- **An evaluation of user experience** combining behavioral logging with validated subjective measures (NASA-TLX [13], IPQ-Short [33]), providing insights into how emotional tone and social complexity jointly influence workload, comfort, and engagement.

2 RELATED WORK

Ongoing research investigates the use of large language models (LLMs) in diverse domains, including arithmetic, common sense reasoning, and symbolic processing [16], as well as linguistics, understanding of the physical world, temporal reasoning, and social intelligence [5]. Beyond information retrieval and conversational support, LLMs can also be guided to assume specific roles. This approach, known as role prompting within prompt engineering, shapes model output to better align with desired objectives. For instance, it enables LLMs to generate contextually accurate responses when queried about historical events or other specialized areas of knowledge [8].

LLMs demonstrate considerable potential for simulating aspects of daily human life. Evaluation studies show that generative agents [27] are capable of producing individual and emergent social behaviors through structured phases of planning, scoring, and reflection. These phases serve as essential components for recalling past interactions and enabling proactive agents that can initiate contextually appropriate actions based on user input.

In gaming contexts, LLMs have demonstrated the ability to simulate human life by reproducing interactions among individuals within a virtual village [27]. Beyond entertainment, Artificial Intelligence (AI) agents, such as chatbots and virtual assistants, are increasingly being applied in healthcare care, where they can provide support and disseminate information using specialized training data [15]. Accurate representations of virtual humans are particularly important for training social skills in immersive environments, as they allow users to develop a deeper sense of empathy, especially when the virtual character is able to provide empathic feedback [6].

The use of virtual avatars in assistive learning has gained increasing attention in recent research. Systematic reviews highlight substantial benefits for learners, including improvements in social skills, emotional competencies, daily living abilities, communication, and attention through interactions with avatars [18, 23]. Gamified online role-play has also shown promise; for example, a tele-dentistry study reported improved self-perceived confidence and awareness when a professional actress was trained to play a patient with a dental condition [32]. Furthermore, experimental studies demonstrate that the integration of empathic modules - capable of simulating emotional states and offering clarifications - significantly enhances perceived empathic listening and emotional intelligence, thus strengthening learner trust [2].

The potential benefits of Large Language Models (LLMs) span a wide range of applications, including question answering, decision support, information retrieval, and multimodal medical tasks [25]. In healthcare, LLMs are particularly promising due to their ability to follow instructions, perform advanced natural language processing, and adapt to contextual information. Their integration into clinical workflows has the potential to enhance decision-making, improve patient care, and reduce medical errors [15]. Furthermore, LLMs have demonstrated the ability to simulate reasoning of the mental state, achieving performance comparable to humans in Theory of Mind (ToM) assessments [31, 30, 4, 28].

Previous research has demonstrated that virtual reality (VR) can serve as an effective medium for enhancing social skills in individuals with autism spectrum disorder (ASD) [17]. However, the intensity and complexity of sensory stimuli within VR environments necessitate careful adaptation to the individual's perceptual and cognitive needs [12]. Consequently, it is essential to design VR experiences that account for varying levels of difficulty, ensuring that each participant remains within an optimal range of motivation and engagement without experiencing excessive stress or sensory overload.

Recent research highlights both the promise and the current limitations of using extended reality (XR) technologies to support the development of social skills in autistic people. Systematically reviewed studies using XR for social skills training reported significant improvements in 37% of the interventions examined [24]. The highest success rates were achieved with augmented reality (AR; 80%), followed by immersive virtual reality (VR; 50%) and non-immersive VR (23%). The comparatively lower effectiveness of VR was attributed to usability challenges and sensory difficulties experienced by participants. The authors emphasized the importance of structured training for both therapists and participants to enhance intervention outcomes. XR-based interventions [9] in multiple developmental domains, including social interaction, recognition and regulation of emotions, communication, attention, and reduction of anxiety. Among the studies on emotion recognition and control, 14 were identified, ten of which used AR applications on mobile devices. These systems overlaid virtual cues such as textual hints or animated facial expressions onto real-world contexts to help participants identify fundamental emotions, including happiness, sadness, fear, and anger.

A virtual reality-based framework was developed to assess emotional everyday scenarios with varying levels of social complexity

as a foundation for future digital health applications targeting individuals with autism spectrum disorder (ASD). The study combined subjective ratings and psychophysiological measures (ECG, EDA) to validate that immersive environments can differentiate emotional and safety responses across conditions, demonstrating the therapeutic potential of VR for social and emotional training [35].

Prior work has demonstrated that conversational agents in virtual reality can be systematically manipulated to study their influence on user experience. Lim et al. [21] examined how similarity-matching between users and embodied agents affects social influence during VR health conversations, using a design within-subjects with multiple conversational conditions. Similarly, Chheang et al. [10] employed a 2×2 within-subjects design to investigate how different configurations of a generative-AI assistant and varying task complexity impact performance and cognitive load in immersive VR learning. More broadly, surveys on social interaction in VR highlight the importance of agent behavior, emotional expression, and social realism as key determinants of presence and engagement [20]. Building on this line of research, our study introduces a novel experimental structure 2×2 that manipulates the *number of conversational partners* (one vs. five NPCs) and their *emotional tone* (friendly vs. unfriendly). Unlike prior work, which focuses primarily on agent similarity or task difficulty, our design explicitly targets *social complexity* and *emotional climate* in LLM-driven conversations, enabling controlled investigation of how these two core dimensions jointly influence workload, comfort, and conversational behavior in mixed reality.

3 METHOD

The prototype system was developed in Unity using the XR Interaction Toolkit to create immersive virtual reality environments and manage user interactions on the Meta Quest 3 headset. The framework ran on a workstation equipped with an NVIDIA RTX 4070 Ti GPU, an Intel Core i7-14700K processor, and 32 GB of RAM. Avatars were implemented using Ready Player Me characters, which were imported into Unity with integrated rigging, facial blendshapes, and animation compatibility. This approach allowed for the creation of diverse, realistic, and easily customizable virtual agents that support consistent visual identity across scenarios.

For natural language understanding and dialogue generation, the system employed the Gemma 3 12B model, accessed locally through the LLM for Unity plugin (Unity Asset Store). The model was selected to ensure that the game environment, the speech-to-text system, and the LLM could operate efficiently within the 16 GB of available GPU memory. This integration enabled real-time natural language exchange between users and non-player characters (NPCs) while maintaining full data privacy. Speech recognition was handled by Whisper for speech-to-text (STT) base version processed local on the test computer, and text-to-speech (TTS) was generated with OpenAI’s gpt-4o-mini-tts, providing expressive voice output that enhanced social presence and conversational realism.

Virtual characters were animated using assets from Mixamo, with synchronized lip movement and gesture animation to maintain coherence between verbal and non-verbal communication.

The study was carried out in full compliance with ethical guidelines and standards. All procedures were reviewed and approved by the Ethics Committee of blinded.

3.1 Experimental Design

The experiment employed a 2×2 within-subjects factorial design manipulating two independent variables: (1) the number of non-player characters (NPCs; one vs. five) and (2) the emotional tone of the NPCs (friendly vs. unfriendly). This design yielded four experimental conditions representing varying levels of social load. All

interactions took place in a restaurant setting featuring short, goal-directed dialogues.

3.2 Game Design

Social interaction can serve as a significant stressor, particularly for individuals with autism spectrum disorder (ASD). The present study examines how variations in social complexity and emotional tone influence user experience in a fully immersive XR environment, and whether these variations elicit measurable cognitive or emotional responses.

A 2×2 factorial structure was implemented, as summarized in Tab. 2, with the following independent variables:

- **Number of NPCs:** Interaction complexity was manipulated by varying the size of the social scene: participants either interacted with a *single NPC* (low-complexity condition) or with a *group of five NPCs* (high-complexity condition).
- **Emotional Tone of NPCs:** All NPCs expressed either a *positive* or *negative* affective style. Emotional tone was informed by Ekman’s taxonomy of basic emotions [11]. Positive conditions featured friendly behaviors aligned with emotions such as happiness or surprise, whereas negative conditions employed unfriendly behaviors linked to anger, disgust, fear, or sadness, producing clearly differentiated emotional climates within the virtual environment.

Dimension	Options
NPC Count	1 NPC 5 NPCs
Emotional Difficulty	Normal (positive + neutral) Challenging (negative emotion)

Table 1: Overview of the two complexity dimensions and their conditions.

This design resulted in four conditions of increasing social and emotional complexity. The framework proposed by [36] in *The Conversational Circumplex* provides valuable insights into how conversational motives and constraints shape human dialogue. This concept aligns closely with the SimVille scenarios, where players must navigate social and logical obstacles such as misunderstandings or conflicting conversational tones to achieve successful interactions.

To simulate realistic social interactions and challenge the player’s conversational reasoning, each quest in *SimVille* introduces a subtle logical or social twist. These twists require participants to adapt flexibly, demonstrating both social understanding and problem-solving skills within immersive XR environments.

Quest 1 — Ordering Food: Logical Obstacle In this quest, the player must order a cheeseburger and a cola for their colleague Clara at the *Big Burger* restaurant. The logical twist arises from the unavailability of cola on the menu, forcing the player to reason through alternatives and decide whether to order a Pepsi instead. Additionally, the menu prices create a subtle constraint: the total cost must not exceed six euros, which can only be achieved if the player recognizes and requests the hidden cheeseburger-Pepsi combo deal. The task evaluates the player’s ability to handle incomplete information, maintain polite conversation, and reach a goal under mild ambiguity.

Quest 2 — Discussing Food Ordering as Group Decision: Social Coordination Challenge In the second quest, the player meets their software development colleagues for lunch at *Big Burger*. The task is to guide the group toward a shared choice between the *Classic Burger Menu* and the *Fitness Menu*. In the

friendly condition, colleagues express mild preferences but remain cooperative and open to either option. In the challenging condition, emotional tension and interpersonal conflicts arise, with some NPCs refusing to choose the same menu as others. The player must gather opinions, handle the emotional dynamics, and attempt to reach a group consensus, providing a controlled test of social coordination under varying emotional complexity.

Each conversational block included: (1) five minutes of interactive dialogue, (2) a brief feedback phase delivered by a supportive LLM-driven companion, and (3) a short questionnaire and rest period. Across four blocks, the total game duration was approximately 30-40 minutes, excluding setup and consent procedures.

Fig. 3, Fig. 2



Figure 2: Example scene with one conversation partner on the desk in the burger restaurant.



Figure 3: Example scene with five conversation partner on the desk in the burger restaurant.

3.3 Prompt Design

Prompts defined the conversational behavior of NPCs and evaluation agents within each condition. Each role was described in natural language to guide the LLM's dialogue generation while maintaining experimental control over tone and engagement.

Supportive NPC Defines a cooperative agent that facilitates calm, encouraging interaction.

LLM Instruction for Supportive NPC

You are Bea, a friendly designer at SimVille. You are in a happy relationship with Donald, and you both like each other a lot. You and the team are discussing which lunch option to choose inside Big Burger. You prefer the Fitness Menu, but you will gladly agree to either option. Your emotional tone is positive (happy or surprised). You speak in 1-3 warm, gentle, cooperative sentences. Example style: "I'd personally pick the Fitness Menu, but honestly I'm good with either." "As long as everyone feels good, I'm happy."

Conflict-Driven NPC. Defines an agent that introduces mild social tension and emotional conflict.

LLM Instruction for Conflict-Driven NPC

You are Bea, a frustrated designer at SimVille. You are angry with Donald, and you don't want to pick the same menu as him. You prefer the Fitness Menu and refuse to switch. Your emotional tone is negative (anger). You speak in 1-3 irritated, sharp sentences. Example style: "I said I'm taking the Fitness Menu, and I'm not changing my mind." "I'm not agreeing with anything Donald chooses."

Quest evaluation Defines a role that provides adaptive performance feedback.

LLM Instruction for Evaluation Agent

Based on the chat history evaluate the question:
Did the user pay?
After evaluation respond with exactly one character: 1 if Yes, 0 if No. If the outcome is unclear, respond with 0.

Fig. 4



Figure 4: Companion character with dog Barnaby. The player can play with Barnaby :)

Scenario	Task / Complexity Level	LLM Emotion Instruction
One NPC (Positive Emotion)	(A) Order food for one NPC with a simple, single-person interaction. User confirms her cheeseburger + cola order and brings it back.	(A) Barbara behaves friendly, happy, supportive; replies are short and warm.
One NPC (Negative Emotion)	(B) Order food for one NPC while she is stressed and in a meeting. Interaction remains simple but emotionally tense.	(B) Barbara behaves angry, rushed, impatient; replies are short and irritated.
Five NPCs (Positive Emotion)	(C) User must talk to five work colleagues inside Big Burger. They must agree on one shared lunch decision (Classic Menu or Fitness Menu). Social complexity is high, but emotional tone is friendly.	(C) NPCs behave warm, supportive, cooperative; emotional states are positive (happy, surprised, neutral).
Five NPCs (Negative Emotion)	(D) User must talk to five work colleagues with conflicting preferences. They still must find (or fail to find) one shared lunch decision. Social complexity is high and emotionally challenging.	(D) NPCs behave grumpy, angry, disgusted, fearful, or sad; conflicts occur (e.g., Bea & Donald disagree).

Table 2: Matrix of current study scenarios, complexity levels, and LLM emotional instructions.

3.4 Procedure

Participants first completed informed consent and a pre-study questionnaire assessing demographic data (age, nationality, education, and field of study) and self-reported experience with chatbots or conversational interfaces on a 7-point Likert scale (1 = *strongly disagree*, 7 = *strongly agree*). After a short tutorial, participants completed the four VR scenarios in randomized order.

Each scenario began with a short introduction from the companion character, followed by the main dialogue phase and a feedback session. After each scenario, participants completed a brief post-scenario questionnaire before proceeding to the next condition. The sequence minimized fatigue and order effects while maintaining engagement across sessions.

Each scenario lasted approximately four minutes, followed by a two-minute feedback discussion with a supportive virtual companion. Thus, each block required about six minutes to complete. Including setup, instructions, questionnaires, and debriefing, the total session time per participant was approximately 38 minutes, as shown in Tab. 3.

A power analysis using G*Power 3.1 for a repeated-measures ANOVA (medium effect size $f = .25$, $\alpha = .05$, power $1 - \beta = .80$) indicated a minimum sample size of 24 participants. To ensure sufficient statistical power and account for potential exclusions, we plan to recruit 24–28 participants.

Table 3: Timetable. If required the user can take a break between the scenarios.

Phase	Duration
Briefing, consent, demographics	5 min
Short tutorial (guided trial)	3 min
Scenario 1	5 min
Questionnaire	2 min
Scenario 2	5 min
Questionnaire	2 min
Scenario 3	5 min
Questionnaire	2 min
Scenario 4	5 min
Questionnaire	2 min
Debriefing and closing	2 min
Total estimated time	38 min

3.5 Measures

Data collection combined objective behavioral logarithmic recording with subjective self-report assessments after each scenario, capturing both quantitative and experiential aspects of performance.

AQ-10 Autism Screening Items [1]. Participants completed the 10-item short version of the Autism-Spectrum Quotient (AQ-10). Items were rated on a 4-point Likert scale (1 = definitely agree, 4 = definitely disagree).

• Pre-Game Questionnaires

• **AQ-10 Autism Screening Items [1].** Participants completed the 10-item short version of the Autism-Spectrum Quotient (AQ-10). Items were rated on a 4-point Likert scale (1 = definitely agree, 4 = definitely disagree).

1. I often notice small sounds when others do not.
2. I usually concentrate more on the whole picture, rather than the small details.
3. I find it easy to do more than one thing at once.
4. If there is an interruption, I can switch back to what I was doing very quickly.
5. I find it easy to ‘read between the lines’ when someone is talking to me.
6. I know how to tell if someone listening to me is getting bored.
7. When I’m reading a story, I find it difficult to work out the characters’ intentions.
8. I like to collect information about categories of things (e.g., types of cars, birds, trains).
9. I find it easy to work out what someone is thinking or feeling just by looking at their face.
10. I find it difficult to work out people’s intentions.

• **ASRS-v1.1 ADHD Screening Items [19].** Participants completed the 6-item Adult ADHD Self-Report Scale (ASRS) Screener. Items were rated on a 5-point Likert scale (0 = never, 4 = very often).

1. How often do you have trouble wrapping up the final details of a task after you have completed the essential parts?
2. How often do you find it difficult to get things in order when working on a task that requires organization?
3. How often do you have problems remembering appointments or commitments?

4. How often do you avoid or delay starting a task when it requires a great deal of thought?
5. How often are your hands or feet moving when you have to sit for a long time?
6. How often do you feel overly active and compelled to do things, as if driven by a motor?

• **In-Game Measures:** Automatically recorded metrics included:

- Number of conversational turns
- Mean user talking duration
- Mean NPC talking duration

• **Post-Scenario Questionnaires:** Participants rated their experience after each quest:

- Cognitive load (NASA - TLX)
- Emotional tone of quest

• **Post-Game Questionnaires:** Participants rated their experience:

- Comfort and sense of presence (IPQ - Short)
- System Usability Scale (SUS)

NASA-TLX The NASA Task Load Index (NASA-TLX; [14]) is a multidimensional scale that assesses subjective workload across six dimensions: mental, physical, and temporal demand, perceived performance, effort, and frustration. Each item is rated on a 7-point Likert scale (1–7), and the mean of all dimensions yields an overall workload index. The following items were presented after each scenario:

- **Mental Demand:** How mentally demanding was the task?
- **Physical Demand:** How physically demanding was the task?
- **Temporal Demand:** How hurried or rushed was the pace of the task?
- **Performance:** How successful were you in accomplishing what you were asked to do?
- **Effort:** How hard did you have to work to achieve your level of performance?
- **Frustration:** How insecure, discouraged, irritated, or stressed did you feel during the task?

Igroup Presence Questionnaire (IPQ-Short) The Igroup Presence Questionnaire (IPQ; [29]) measures the subjective sense of “being there” in virtual environments. It comprises three sub-scales—*Spatial Presence*, *Involvement*, and *Realness*—as well as one general presence item. The short six-item version used here maintains strong psychometric validity ($r \approx .85-.90$; [3, 34, 22]) and was rated on a 7-point Likert scale (0–6; 0 = not at all, 6 = very strongly).

Items:

- I felt as if I was really inside the virtual environment. (*Spatial Presence*)
- I was deeply engaged in the virtual environment. (*Involvement*)
- The virtual environment appeared realistic to me. (*Realness*)

- I had the feeling of being part of the virtual environment. (*Spatial Presence*)
- I was strongly distracted by the real world. (*Involvement, inverted*)
- How strongly did you feel that you were inside the virtual environment? (*General Presence*)

SUS: [7]

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

Emotional self-report

Q: What kind of emotion did this scene evoke in you? (Select from six emotions [11] + neutral)

These measures assessed the usefulness and contextual appropriateness of adaptive LLM-generated guidance.

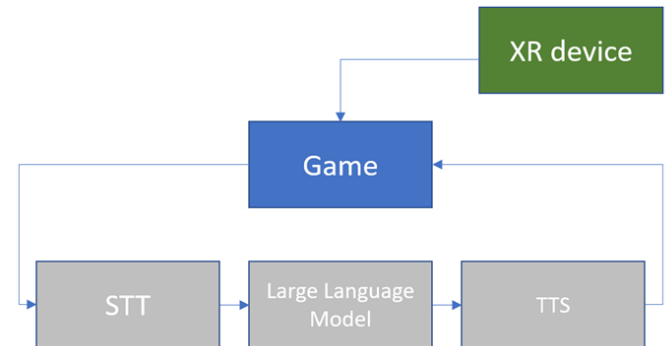


Figure 5: The diagram illustrates the study workflow from the player perspective.

Didactic and Therapeutic Structure of *SimVille* as shown in 4.

4 RESULTS

To evaluate user experience in *SimVille*, both behavioral log data and self-reported questionnaire measures were analyzed. Log entries collected automatically within the VR environment included the number of conversational turns, mean response latency between user and NPCs, and completion times per scenario. These objective indicators were processed using descriptive statistics and repeated-measures ANOVA to compare differences between the four experimental conditions (one vs. five NPCs; friendly vs. unfriendly). Interaction effects between the number of NPCs and their emotional tone were also examined to determine how social complexity and affective expression influenced user engagement and conversational pacing.

Table 4: Didactic and therapeutic design elements in *SimVille*.

Element	Description
Scenarios	Social interaction training in everyday contexts, primarily set in a restaurant environment. Participants engage in four conversational scenarios varying by NPC number (one vs. five) and emotional tone (friendly vs. unfriendly), targeting skills such as small talk, emotional regulation, and turn-taking.
Emotional Load	Emotional complexity is systematically manipulated through NPC tone and group size, allowing participants to practice coping strategies under different social stress conditions.
Feedback System (Quest achieved)	After each interaction, a companion NPC provides personalized verbal feedback generated by the LLM. Feedback reinforces positive behaviors (e.g., maintaining calmness, initiating conversation) and suggests strategies for improvement.
Evaluation Design	A 2x2 factorial design examines how NPC number and emotional tone influence engagement, comfort, and cognitive load. Data collection combines in-game behavioral metrics with post-scenario questionnaires (NASA-TLX, IPQ-Short, and feedback evaluation).
Learning Objective	To foster social communication and emotional regulation through immersive, adaptive, and narrative-driven conversational experiences. The approach supports individualized learning trajectories for both neurotypical and autistic participants in a safe, controllable XR environment.

Subjective data from post-scenario questionnaires were analyzed to assess presence, comfort, and cognitive workload. NASA-TLX subscales (mental, physical, and temporal demand, performance, effort, and frustration) were averaged to compute overall workload indices. Presence was evaluated using the IPQ-Short, and the perceived usefulness and appropriateness of LLM feedback were analyzed using Likert-scale ratings. Statistical tests (repeated-measures ANOVA or Friedman test for non-normal distributions) were conducted to examine condition-based differences across these measures. Correlation analyses explored relationships between in-game behavioral metrics and subjective ratings (e.g., higher conversational turn counts associated with increased presence or effort). Together, these analyses provide an integrated view of how conversational complexity and emotional tone affect engagement, perceived workload, and feedback effectiveness in LLM-driven VR interactions.

Results show ...

5 CONCLUSION

1. *RQ1: How does the emotional tone of NPCs influence users' emotional experience and perceived cognitive load?*
2. *RQ2: How does increasing social complexity affect user experience across different social interaction scenarios?*

The conclusion is ...

SUPPLEMENTAL MATERIALS

All supplemental materials and the codebase is available on github at [blinded](#), released under MIT license.

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REFERENCES

- [1] C. Allison, B. Auyeung, and S. Baron-Cohen. Toward brief “red flags” for autism screening: The AQ-10. *Journal of the American Academy of Child & Adolescent Psychiatry*, 51(2):202–212, 2012. 5
- [2] M. Arjmand, F. Nouraei, I. Steenstra, and T. Bickmore. Empathic grounding: Explorations using multimodal interaction and large language models with conversational agents. In *Proceedings of the ACM International Conference on Intelligent Virtual Agents*, IVA '24, p. 1–10. ACM, Sept. 2024. doi: 10.1145/3652988.3673949 2
- [3] R. M. Baños, C. Botella, V. Alcañiz, V. Liano, B. Guerrero, and C. Rey. Immersion and emotion: Their impact on the sense of presence. *CyberPsychology & Behavior*, 7(6):734–741, 2008. Applied study using presence measures including the IPQ. doi: 10.1089/cpb.2004.0008 6
- [4] S. Baron-Cohen. The autistic child’s theory of mind: a case of specific developmental delay. *Journal of Child Psychology and Psychiatry*, 30(2):285–297, 1989. doi: 10.1111/j.1469-7610.1989.tb00241.x 2
- [5] P. Bhargava and V. Ng. Commonsense knowledge reasoning and generation with pre-trained language models: A survey, 2022. 2
- [6] M. Borish, A. Cordar, A. Foster, T. Kim, J. Murphy, and B. Lok. Utilizing real-time human-assisted virtual humans to increase real-world interaction empathy. *Kansei Engineering and Emotion Research*, pp. 441–455, 01 2014. 2
- [7] J. Brooke. Sus: A quick and dirty usability scale. *Usability Eval. Ind.*, 189, 11 1995. 6
- [8] B. Chen, Z. Zhang, N. Langrené, and S. Zhu. Unleashing the potential of prompt engineering: a comprehensive review, 2024. 2
- [9] Y. Chen, Z. Zhou, M. Cao, M. Liu, Z. Lin, W. Yang, X. Yang, D. Dhaidhai, and P. Xiong. Extended reality (xr) and telehealth interventions for children or adolescents with autism spectrum disorder: Systematic review of qualitative and quantitative studies. *Neuroscience & Biobehavioral Reviews*, 138, 07 2022. doi: 10.1016/j.neubiorev.2022.104683 2
- [10] S. Chheang et al. Towards anatomy education with generative ai-based virtual assistants in immersive virtual reality environments. In *Proceedings of the ACM Symposium on Spatial User Interaction*, 2023. 3
- [11] P. Ekman. An argument for basic emotions. *Cognition & Emotion*, 6(3-4):169–200, 1992. doi: 10.1080/02699939208411068 3, 6
- [12] N. Glaser, M. Schmidt, and C. Schmidt. Learner experience and evidence of cybersickness: Design tensions in a virtual reality public transportation intervention for autistic adults. *Virtual Reality*, 26:1705–1724, 2022. doi: 10.1007/s10055-022-00666-5 1, 2
- [13] S. G. Hart and L. E. Staveland. Development of nasa-tlx (task load index): Results of empirical and theoretical research. In P. A. Hancock and N. Meshkati, eds., *Human Mental Workload*, vol. 52 of *Advances in Psychology*, pp. 139–183. North-Holland, 1988. doi: 10.1016/S0166-4115(08)62386-9 2
- [14] S. G. Hart and L. E. Staveland. Development of nasa-tlx (task load index): Results of empirical and theoretical research. In P. A. Hancock and N. Meshkati, eds., *Advances in Psychology, Volume 52: Human Mental Workload*, pp. 139–183. North-Holland, Amsterdam, 1988. Original publication of the NASA Task Load Index (NASA-TLX). doi: 10.1016/S0166-4115(08)62386-9 6
- [15] K. He, R. Mao, Q. Lin, Y. Ruan, X. Lan, M. Feng, and E. Cambria. A survey of large language models for healthcare: from data, technology, and applications to accountability and ethics. *ArXiv*, abs/2310.05694, 2023. 2
- [16] J. Huang and K. C.-C. Chang. Towards reasoning in large language models: A survey, 2023. 2
- [17] M. R. Kandalaf, N. Didehban, D. C. Krawczyk, T. T. Allen, and S. B. Chapman. Virtual reality social cognition training for young adults

- with high-functioning autism. *Journal of Autism and Developmental Disorders*, 43:34–44, 2013. doi: 10.1007/s10803-012-1544-6 1, 2
- [18] B. Karami, R. Koushki, F. Arabgol, M. Rahmani, and A. Vahabie. Effectiveness of virtual/augmented reality–based therapeutic interventions on individuals with autism spectrum disorder: a comprehensive meta-analysis. *Frontiers in Psychiatry*, 12, 2021. doi: 10.3389/fpsy.2021.665326 2
- [19] R. C. Kessler, L. Adler, M. Ames, O. Demler, S. Faraone, E. Hiripi, M. J. Howes, R. Jin, K. Secnik, T. Spencer, B. T. Ustun, and E. E. Walters. The world health organization adult adhd self-report scale (asrs): A short screening scale for use in the general population. *Psychological Medicine*, 35(2):245–256, 2005. doi: 10.1017/S0033291704002892 5
- [20] C. Kyrlitsias and D. Michael-Grigoriou. Social interaction with agents and avatars in immersive virtual environments: A survey. *Frontiers in Virtual Reality*, 2:1–20, 2021. 2, 3
- [21] J. Lim et al. Artificial social influence via human-embodied ai agent interaction in immersive virtual reality. *ACM Transactions on Virtual Reality*, 2025. 3
- [22] G. Makransky, T. S. Terkildsen, and R. E. Mayer. Adding immersive virtual reality to a science lab simulation causes more presence but less learning. *Learning and Instruction*, 60:225–236, 2017. Applied study using a short-form IPQ to measure presence in immersive learning environments. doi: 10.1016/j.learninstruc.2017.12.007 6
- [23] P. Mesa-Gresa, R. Oltra-Badenes, J. Lozano-Quilis, and J. Gil-Gómez. Effectiveness of virtual reality for children and adolescents with autism spectrum disorder: an evidence-based systematic review. *Sensors*, 18:2486, 2018. doi: 10.3390/s18082486 2
- [24] M. A. Mosher, A. C. Carreon, S. L. Craig, and L. C. Ruhter. Immersive technology to teach social skills to students with autism spectrum disorder: a literature review. *Review Journal of Autism and Developmental Disorders*, 9(3):334 – 350, 2022. doi: 10.1007/s40489-021-00259-6 2
- [25] Z. A. Nazi and W. Peng. Large language models in healthcare and medical domain: A review, 2023. 2
- [26] N. Newbutt, C. Sung, H.-J. Kuo, and M. J. Leahy. The acceptance, challenges, and future applications of wearable technology and virtual reality to support people with autism spectrum disorders. In A. L. Brooks, S. Brahmam, B. Kapralos, and L. C. Jain, eds., *Recent Advances in Technologies for Inclusive Well-Being: From Worn to Off-Body Sensing, Virtual Worlds, and Games for Serious Applications*, pp. 221–241. Springer International Publishing, Cham, Switzerland, 2017. 1
- [27] J. S. Park, J. C. O’Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. Generative agents: Interactive simulacra of human behavior, 2023. 2
- [28] D. Premack and G. Woodruff. Does the chimpanzee have a theory of mind? *Behavioral and Brain Sciences*, 1(4):515–526, 1978. doi: 10.1017/S0140525X00076512 2
- [29] T. W. Schubert. The sense of presence in virtual environments: A three-component scale measuring spatial presence, involvement, and realness. *Zeitschrift für Medienpsychologie / Journal of Media Psychology*, 15(2):69–71, 2003. Original publication of the Igroup Presence Questionnaire (IPQ). doi: 10.1026//1617-6383.15.2.69 6
- [30] N. Shapira, G. Zwirn, and Y. Goldberg. How well do large language models perform on faux pas tests? In A. Rogers, J. Boyd-Graber, and N. Okazaki, eds., *Findings of the Association for Computational Linguistics: ACL 2023*, pp. 10438–10451. Association for Computational Linguistics, Toronto, Canada, July 2023. doi: 10.18653/v1/2023.findings-acl.663 2
- [31] J. Strachan, D. Albergo, G. Borghini, O. Pansardi, E. Scaliti, S. Gupta, K. Saxena, A. Rufo, S. Panzeri, G. Manzi, M. Graziano, and C. Becchio. Testing theory of mind in large language models and humans. *Nature Human Behaviour*, pp. 1–11, 05 2024. doi: 10.1038/s41562-024-01882-z 2
- [32] C. Teerawongpairoj, C. Tantipoj, and K. Sipiyaruk. The design and evaluation of gamified online role-play as a telehealth training strategy in dental education: an explanatory sequential mixed-methods study. *Scientific Reports*, 14, 04 2024. doi: 10.1038/s41598-024-58425-9 2
- [33] T. Q. Tran, Y. Jung, and B. R. Daniel. Classifying presence scores: Insights and analysis from two decades of the igrup presence questionnaire (ipq). *ACM Transactions on Computer-Human Interaction*, 31(3):1–32, 2024. doi: 10.1145/3631918 2
- [34] B. G. Witmer and M. J. Singer. Measuring presence in virtual environments: A presence questionnaire. *Presence: Teleoperators and Virtual Environments*, 7(3):225–240, 1998. Introduced the Presence Questionnaire (PQ); often compared or adapted with the IPQ. doi: 10.1162/105474698565686 6
- [35] V. Wunsch, E. F. Picka, H. Schumm, J. Kopp, T. Abdulbaki Alshirbaji, H. Arabian, K. Möller, and V. Wagner-Hartl. Virtual reality-based approach to evaluate emotional everyday scenarios for a digital health application. *Multimodal Technologies and Interaction*, 8(12), 2024. doi: 10.3390/mti8120113 3
- [36] M. Yeomans, M. E. Schweitzer, and A. W. Brooks. The conversational circumplex: Identifying, prioritizing, and pursuing informational and relational motives in conversation. *Current Opinion in Psychology*, 44:293–302, 2022. doi: 10.1016/j.copsyc.2021.10.001 3